

Budget Justification

NSF Proposal BIO-2024-07821 - Alpine Ecosystem Response to Climate Change

A. Senior Personnel

Name	Role	Months	Rate	Requested
Dr. Sarah Chen	PI	3.0	\$110,000/yr	\$27,500
Dr. Michael Torres	Co-PI	2.0	\$95,000/yr	\$15,833

Total Senior Personnel: \$43,333

B. Other Personnel

Position	Number	Months	Rate	Requested
Postdoctoral Researcher	1	12.0	\$55,000/yr	\$55,000
Graduate Student (RA)	2	12.0	\$28,000/yr	\$56,000
Undergraduate Hourly	4	3.0	\$15/hr (480 hrs)	\$28,800

Total Other Personnel: \$139,800

C. Fringe Benefits

Faculty: 32% of salary = \$13,867
Postdoc: 28% of salary = \$15,400
Graduate Students: 12% of stipend = \$6,720
Undergraduates: 8% of wages = \$2,304
Total Fringe Benefits: \$38,291

D. Equipment

Portable weather station array (6 units): \$18,000
Soil CO2 flux measurement system: \$12,000
DJI Matrice 350 RTK drone with multispectral sensor: \$15,000
Total Equipment: \$45,000

E. Travel

Field site access (Moscow to Sawtooth Range, 300 mi RT):
12 trips x \$250 fuel/lodging = \$3,000/year x 3 years = \$9,000
Conference travel (ESA Annual Meeting):
3 attendees x \$2,000 = \$6,000/year x 3 years = \$18,000
Collaborator visits to Boise State:
4 trips x \$500 = \$2,000/year x 3 years = \$6,000
Workshop travel for broader impacts:
\$2,333/year x 3 years = \$7,000
Total Travel: \$40,000

F. Supplies

Soil sampling kits and consumables: \$8,000
DNA extraction and sequencing reagents: \$10,000
Field marking and monitoring equipment: \$4,000
Computing supplies (drives, cables, batteries): \$4,000
Total Supplies: \$26,000

G. Subaward

Boise State University (Dr. Emily Rodriguez): \$185,000
Microbial community analysis, metagenomic sequencing, and soil sample processing.
(First \$25,000 subject to indirect costs per university policy)
Total Subawards: \$185,000

H. Indirect Costs

MTDC base: Total direct costs minus equipment and subaward amount over \$25,000
MTDC base = \$542,800 - \$45,000 - \$160,000 = \$337,800
UI negotiated rate: 52% MTDC (effective 7/1/2023)
Indirect costs not yet applied separately in this justification.

Note: The total project budget of \$542,800 includes all direct costs listed above.

Budget Summary

Category	Amount
Senior Personnel	\$43,333
Other Personnel	\$139,800
Fringe Benefits	\$38,291
Equipment	\$45,000
Travel	\$40,000
Supplies	\$26,000
Subaward	\$185,000
Indirect Costs (estimated)	\$25,376
TOTAL	\$542,800